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ORIGINAL ARTICLE

A novel data-driven approach to detect and predict customer transitions in the marketing funnel

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Abstract
This study braids together insights from two distinct areas of research: customer engagement within brands and social data analytics. Leveraging a proprietary B2B dataset from LinkedIn comprising over 200 million advertiser interactions over a subset of de-identified members, we present a data-driven exploration of the dynamics of the so-called marketing funnel. By bridging the gap between real-world digital metrics and theoretical marketing funnel models, we offer empirical insights into the interplay between engagement types and customer transitions towards purchase. Our analysis delineates the structural underpinnings of the derived marketing funnel to shed light on the efficacy of various engagement strategies in propelling users towards deeper funnel stages, aiding advertisers in predicting and targeting engagements effectively. We make three contributions: first, we propose a novel data-driven methodology to create a representation of the marketing funnel. Second, we apply the data-driven marketing funnel model on a proprietary dataset from LinkedIn advertisements with a subsequent descriptive and predictive analysis of the funnel stages and transitions to showcase the practicability and usefulness of the approach. Third, we analyze the relationship between user engagement and stage transitions within the created marketing funnel, identifying what user engagements are predictive of future stage transitions.

Keywords Marketing funnel · Customer engagement · Predictive analytics · Social data analytics

1 Introduction

Research attention has surged towards causality and explainability in machine learning and artificial intelligence approaches within social e-commerce analytics. This surge is driven by the practical need for precise targeting of potential customers. Social media analytics, now facilitated by no-code dashboards, offer extensive tracking and analysis capabilities to brands and researchers. Notably, early successes such as MySpace's ad campaigns demonstrated significant (28%) return on investment (ROI) (Fodor and Hoffman 2010). Platforms like Meta's Facebook and Microsoft's LinkedIn are examples offering diverse campaign optimization options, reflecting the evolving landscape of social advertising.

Literature presents diverse perspectives on social online advertising effectiveness, from skepticism (Blake et al. 2015; Lewis and Rao 2015), to phased (Hoban and Bucklin 2015) or solid empirical support (Johnson et al. 2017). The "marketing funnel"is the domain application term for the advertisement-to-impact process flow between firms and customers. A data-driven marketing funnel model can resolve the tension around effectiveness of social media advertising by offering insights into campaign effects on user behavior. This study, focusing on LinkedIn, asks and answers: what are the requirements for establishing a flexible, data-driven method for modeling the marketing funnel?

We achieve this first goal of the method-a data-driven solution based on customer behavior and customizable by brand"by depicting the stages of the marketing funnel and its stages as user behavior patterns. The analyses and views are the authors' own and do not represent official positions of LinkedIn or Microsoft. Data use complied with internal review and privacy safeguards. The case study is an observational, non-intervention-based assessment of user

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interactions with advertisement campaigns across users at scale. In total, the study captures over 200 million advertising interactions. The method involves clustering customer behavior into distinct funnel stages, allowing for customer insights and predictive analytics concerning future customer behavior. Our proposed data-driven marketing funnel uses unsupervised learning to cluster customer behavior into distinct stages representing common behavior patterns in the marketing funnel. Additionally, predictive modeling is employed to anticipate customer transitions between stages, aiding in resource allocation and targeting strategies.

The second goal is to improve our understanding of customer movements through the marketing funnel by identifying the antecedent engagements for stage transitions via a predictive model. Generally, engagements precede transitions and might indicate future customer behavior (Paschen et al. 2020). However, the degree to which different engagements serve as a signal that member behavior will change soon is unknown, thus is the basis of the second research question: which signals predict stage transitions? For example, we can model the number of impressions, clicks, shares, and comments required to move customers toward the next stage within the marketing funnel. With the proposed predictive model, we predict when members are ready to transition to the next stage because they have sufficiently engaged within their current stage. Knowing that members are ready to transition stages within the marketing funnel is valuable to the advertiser's strategy as it can serve as a more reliable indicator for resource deployment or constraint. In addition, the ability to predict customer readiness for new engagement in the marketing funnel allows advertisers to better target their members. We infer that better targeting will lead to improved outcomes as understanding which engagements precede transitions strategically places advertisers in influencing customer behavior.

The empirical validation of our approach is firmly rooted in explainability and precision of the measurement to support the reliability and replicability of the proposed approach. Specifically, we link to and extend the theory of the marketing funnel (see Related Literature) by emulating how users move toward the latter stages of engagement (see A Data-driven Approach to Model Marketing Funnels). We utilize over sixty engagement metrics via LinkedIn ad campaigns and analyze them through daily, weekly, and monthly windows over one year (see Case Study: A Data-driven Marketing Funnel for LinkedIn). Building on these engagement types, we use K-Means clustering to derive the marketing funnel stages and decide on a specific number of funnel stages that adequately capture and represent different customer behaviors. In the next step, we use descriptive and predictive analysis techniques to study customer transitions between the marketing funnel stages (see Case Study: Predicting Future Funnel Stage Transitions). We conclude by delineating the practical and theoretical research contributions, pointing out areas of future research.

This paper addresses a specific theoretical gap: related works on the marketing funnel fail to quantify its characteristics, primarily its structure of stages and flow of users between its stages. Rather, most funnel-analytics rely on predefined coarse stages that pertain to business operations rather than data-driven stages based on actual user behaviors. The state of the structure and flow of the marketing funnel in related works provides limited guidance on its practical inner workings. We respond by (i) deriving interpretable stages directly from observed engagement patterns via unsupervised learning and (ii) predicting short-horizon transitions as a function of antecedent engagements. The empirical sections evaluate these behavior-based stages and transition models on large-scale LinkedIn data and, in each case, indicate how the findings speak to this gap: clarifying when stage structure is stable, which engagements are predictive of subsequent movement, and the practical boundary conditions of the approach.

2 Related literature

2.1 The importance of social advertising for brands

The fact that advertising is interlinked with social media is a practically foregone conclusion. Businesses view social media platforms through the potential benefits they offer, and brands view social media campaigns as a method for creating both tangible and intangible firm value that improves business performance (Kumar et al. 2016). This is because people not only see ads but are influenced by them (Kumar et al. 2016), which aligns with the strategic goals of communicating with users, influencing their actions, and building relationships (Swani et al. 2014; Saez-Trumper et al. 2014; Wright et al. 2010; Voorveld et al. 2018). For example, Adidas announced it would focus its marketing efforts exclusively on digital and social channels (Voorveld et al. 2018). Customer brand interactions impact individual-level outcomes regarding customer spending, purchase intentions, brand trust, loyalty, and customer brand engagement (Swani et al. 2014) and a decreased time users need to make a buying decision (Kumar et al. 2016). The interactions are related to firm-level outcomes like sales and advertising effectiveness (Lee and Hong 2016; Wiese et al. 2020), return on investment (Crowe et al. 2024), and stock prices (Swani et al. 2014). By building mutually satisfying relationships, brands earn and retain their business (Wright et al. 2010).

Advertisers face the choice of selecting where they advertise. For instance, Twitter (now X) allows management of own communications channels, permitting platform collaboration and questions between brands and users. An important aspect that advertisers consider when choosing a platform is the targeting option (Saez-Trumper et al. 2014). For example, Facebook does not allow targeting based on a user's popularity or level of activity but includes targeting parameters for location, gender, and interests. Advertising according to the platform's character provides creative outlets where platforms share content and interact with user interests (Voorveld et al. 2018). Some brands provide a higher rate of keeping users surfing on their site, which has been referred to as the platform's rate of distraction, which impacts the influence of social media on consumers (Hughes et al. 2019). For example, Facebook has a higher rate of users becoming distracted than blogs (Hughes et al. 2019).

Social media has been found to impact customer engagement, accelerate the purchase process (Kumar et al. 2016), and generate business engagement on platforms such as Twitter (Swani et al. 2014). Users who take an interest in a social media advert more firmly consume the presented information (Aarts et al. 2012). Further studies document how social media affects customer engagement and the purchase process (Kumar et al. 2016). Social media firm performance largely depends on its ability to affect customer attitudes and behaviors (IEDP Ideas for Leaders (CMO Survey) 2018). Moreover, research demonstrates the successful prediction of consumer behavior based on news, posts from a user's friends, and user interests (Xu et al. 2012).

Advertising has a role in the decision-making process and can improve brand attitudes and generate purchase intent (Chen and Dredze 2018; Lavidge and Steiner 1961; Bruyn and Lilien 2008). However, social media might take time to influence customer purchase behaviors (Kumar et al. 2016). Even something as simple as short public statements on social media can spur transaction activity (Nielsen 2012). Greater awareness can lead toward loyalty (Colicev et al. 2019; Oliver 1999).

2.2 Types of content in online advertising

Empirical work shows social media ads influence spending, loyalty, and even stock prices (Kumar et al. 2016; Lee and Hong 2016; Wiese et al. 2020); brands deploy both firm-generated and user-generated content, each affecting purchase intent in different ways (Goh et al. 2013); critical factors are targeting options (Saez-Trumper et al. 2014), platform "distraction rates" (Hughes et al. 2019), and the balance between awareness metrics (impressions, likes) and outcome metrics (clicks, conversions); our funnel model unifies these perspectives by measuring all common engagements on the same behavioural scale.

2.3 Considerations for behavioral social analytics models

Social media users exhibit distinct behavioral patterns when in different stages of the marketing funnel (Jansen and Schuster 2011). For example, the behaviors associated with early branding stages (impressions and post engagement) differ from those in later conversion stages (landing page clicks and conversions). However, behavior patterns consist of more than a single engagement (Hall et al. 2018). For example, customers learning about a brand might like firm posts, visit their company page, and comment on firm posts. The combination of behaviors over time consists of a behavioral pattern.

2.4 Research gap: a data-driven approach to model marketing funnels

To address the challenges with current marketing funnel approaches, we propose a data-driven method to create a marketing funnel from observed user behavior and engagements. Our approach consists of three distinct steps. First, the "Data and Metrics" step, in which the data are collected and the metrics which represent user engagement with the company are selected. Second, the "Data-driven Marketing Funnel" step, where the marketing funnel is created using unsupervised learning. Third, the "Analysis and Prediction" step, where the derived marketing funnel is analyzed, and predictive analytics models are created to predict transitions of users and customers throughout the marketing funnel. Figure 1 shows the general approach and the three overall stages.

3 Data and metrics selection

In the first step of the data-driven marketing funnel approach, we select the data and metrics used to measure interactions and engagement. This is a crucial part of the overall approach as the selected metrics influence the clustering in the subsequent step. Specifically, different selections of metrics will result in different identified funnel stages, thus impacting the overall results from an analysis and prediction perspective as well.

From a general perspective, researchers and practitioners can guide this selection process by identifying which data can be collected or are already available, which metrics in the data best proxy the relevant user behavior of interest, and if differences in user behavior as measured by the

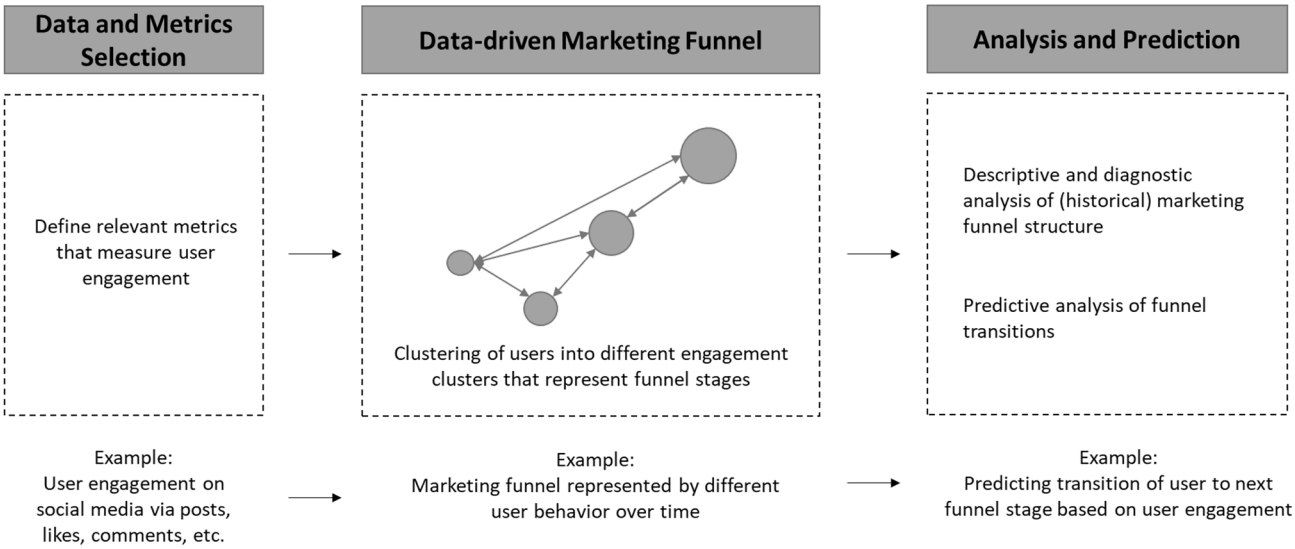


Fig. 1 Data-driven marketing funnel approach

selected metrics can yield actionable insights once the marketing funnel has been created. For example, in our case of social media marketing, available data can be common user interaction types with the social media site such as likes, comments, and clicks. These metrics, and the frequency of the metrics for individual users, quantify different stages of user behavior and engagement with the social media site and can later be used to identify how users transition between different stages.

We decide which user metrics to include in the quantification of user data via unsupervised learning (clustering). The selected metrics directly influence the created marketing funnel. Similar to using clickstream data to measure user behavior in social networks (Wang et al. 2016), we prefer social media metrics that are common across platforms, such as impressions, clicks, comments, and conversions. However, we also explore using less common metrics, such as company page follows, since our methodology is flexible. The metric selection is made in a larger context to ensure our case study is applicable to other use cases.

The data processing performs an analysis concerning which metrics should be included when clustering users based on their behavior as it relates to the marketing funnel. The metrics included in the marketing funnel were derived based on random forest regression model. We trained the random forest to predict user landing page clicks and conversions. We selected columns for this article's clustering approach based on the most performant columns in the random forest. Specifically, we utilized the feature importance from a random forest regression model trained to predict landing page clicks and conversions. Hence, while the metrics selected for the clustering model are biased toward capturing particular engagements, we also validated that metrics were not highly correlated.

4 Generating the marketing funnel

Based on the selected, significant features described in the previous section, we generate a data-driven marketing funnel which allows us to capture major patterns of distinct user behavior. As user behavior changes, we can expect users to move between stages. For instance, a user might move from a "disengaged but interested" stage to a later "frequent interaction" stage. We aim to capture user movement in the marketing funnel described as changes in user behavior. Hence, the funnel stages should be distinct and represent substantially different behaviors. The stages should also be actionable and lead to insights about user behavior in the marketing funnel. By measuring user behavior over specific timeframes, we can track user transitions between the funnel stages. Understanding of how individual users move through the marketing funnel, how likely they are to move to specific stages of the funnel, and what drivers impact these transition probabilities are valuable insights that can be generated by this point of view.

While the overall idea to use clustering, or a similar unsupervised learning approach, to create data-driven funnel stages is straightforward, there are many design decisions in this stage that can impact the resulting marketing funnel. These decisions, represented as parameters in the data-driven approach, are shown in Table 1. Our approach sets itself apart from existing marketing funnel approaches due to its data-driven nature: instead of deciding stages of the marketing funnel based on business practices, such as

Table 1 Design parameters for the data-driven marketing funnel approach

Parameter	Description
Timeframe t	The time period over which individual user behavior is aggregated into a single observation, e.g., this could be a year, a month, a week, or a day.
Number of distinct funnel stages / clusters, k	The number of clusters, or funnel stages, that should be created to differentiate between different user behaviors.
Set of selected metrics to measure user behavior, s	The metrics, or variables, that are used to measure user behavior. As these metrics are used to quantify how similar two observations are and if they should fall into the same funnel stage/cluster, different metric sets generally lead to different allocation of observations to specific funnel stages.

getting leads and calling leads, we use an unsupervised approach and allow the model to decide on the best way to cluster user behavior.

In the previous section, we discussed how metrics can be selected to measure relevant user behavior in a given scenario (set of metrics, s). Next, we need to consider the relevant timeframe for user behavior (Crowe et al. 2024). For instance, we need to choose whether user behavior should be measured daily, weekly, monthly, or yearly? The choice of a specific timeframe will not only affect the number of total observations for the data-driven approach (without loss of generality, we will have more observations when we use shorter timeframes), it will also determine the granularity of the created marketing funnel. For instance, selecting to measure user behavior on a daily basis might be too granular and lead to many cluster transitions where individual users fluctuate between days of significant engagement and being disengaged (e.g., due to lack of time during a workweek), while an aggregation over an entire year might not capture changes in user behavior adequately and in a timely manner. This is a decision that needs to be considered in the context of the available data (i.e., how often are user interactions measured), the appropriateness of different timeframes from a user behavior perspective (i.e., is the focus on short-term changes or long-term changes), and the desired use of the marketing funnel (i.e., should it be used for short-term marketing activities that try to influence user behavior, or long-term user/customer development).

Finally, from an unsupervised learning perspective, we also need to decide how many distinct clusters we want to create. This, again, is a design decision that needs to be made for the specific scenario and desired granularity of the funnel (Delattre and Hansen 1980). A smaller number of clusters might not adequately distinguish between distinct user behaviors, while a larger number of clusters might lead to a funnel with an insufficient number of users. An appropriate number of clusters should balance the distinction between distinct user behavior with the effort to create actionable strategies/ engagements with individual clusters.

The clusters created also serve as the basis for deriving the distinct stages of the marketing funnel. Cluster analysis has been used to empirically capture data in the B2B customer journey (Marvasti 2021). Specifically, as the clusters capture different member behavior, the clusters can be directly used as a first model of marketing funnel stages. For instance, depending on the metrics selected to measure relevant behavior, clusters will represent users with similar engagement behavior for some or all of the considered metrics. Yet, the clusters might not be the final marketing funnel stages that are created. As clustering builds the different clusters based on quantitative measurements of engagement metrics, it can occur that separate clusters are built for engagement types that are qualitatively similar yet quantitatively distinct. An example would be clusters using the number of comments to measure engagement. The clustering algorithm might identify separate clusters for no comments (no engagement), single/few comments, and many comments within the timeframe, whereas qualitatively the binary distinction between "no comment vs at least one comment" might better capture the engagement that we are interested in. The created clusters should be seen as a first step towards creating the (customized) marketing funnel.

5 Analysis and prediction

Once the data-driven marketing funnel has been created in the previous step, we can now analyze its composition and generate insights that can be used to tailor marketing activities for individual users in specific funnel stages. Generally, we can distinguish between two main types of analysis approaches: descriptive and diagnostic analysis of the created marketing funnel and observed transitions therein, and predictive analysis of individual transitions between funnel stages (Delen and Ram 2018).

From a descriptive and diagnostic perspective, the created marketing funnel can be analyzed to gain a better understanding of previous user behavior and to identify if the desired user behavior is adequately captured by the created marketing funnel. For instance, questions that can be relevant for this perspective are: What distinguishes the different marketing funnel stages? What proportion of users are in specific stages? What are the transition probabilities between the different funnel stages?

Taking a predictive analytics lens, the created marketing funnel can also be used to pro-actively estimate transition probabilities between different funnel stages. For example, we can estimate the transition probability from funnel stage A to another funnel stage B based on the observation's

current position and based on the observed user behavior metrics. Generally, these predictive analyses focus on two distinct goals. First, the analysis of whether the funnel stage transitions can be predicted (and how accurately this can be predicted). Second, from an inference perspective, we identify the factors that influence the probability of a transition. This can be particularly useful if we are interested in identifying actionable information to steer users from one funnel stage to another. For example, predictive models can identify events that increase the likelihood of customers transitioning to a desired funnel stage, hence creating actionable results that can be used to improve customer engagement with a specific topic.

Overall, we can summarize the methodology behind our data-driven marketing funnel approach as follows: First, it is flexible and adaptable to different scenarios as the decision what counts as relevant user behavior is part of its design. Second, different design parameters will lead to potentially different marketing funnel representations, and the right combination of parameters needs to be determined for a given setting. Third, using descriptive and predictive analytics on the created marketing funnel can yield actionable insights to better understand and/or to improve customer interactions. In the next section, we showcase our novel data-driven marketing funnel approach in the context of a specific use case: Creating a data-driven marketing funnel for the social media site LinkedIn.

6 Case study: a data-driven marketing funnel for LinkedIn

To showcase both a concrete implementation of our proposed data-driven marketing funnel approach and the usefulness of this approach to gain insights into user behavior and transitions within the marketing funnel, we now present a case study using a proprietary dataset from LinkedIn, a large international social media site for professional networking and recruiting.

6.1 Data origin and metric selection for user behavior

We use a proprietary dataset collected by LinkedIn, which at the time had data from 2018 through 2021, though we only used 12 months of data due to its size. A subset of advertisers and members were also selected for tracking through the marketing funnel. Therefore, the data is not necessarily representative of LinkedIn or advertisers and cannot be used to infer about the larger LinkedIn advertising ecosystem. All tracking was according to LinkedIn policies, privacy principles, and member consent. The data includes all engagements from the selected LinkedIn users across all selected advertisements from the selected advertisers, which includes a robust and growing global member base across diverse industries and geographies. The dataset utilized for the cluster analysis describes member-advertisement interactions, and the raw data metrics describe LinkedIn member behaviors towards advertisers. The features correspond to the dataset columns and include account id, impressions, landing page clicks, other engagements, external website conversions, external website post-view conversions, external website post-click conversions, likes, comment likes, comments, shares, follows, one-click lead form opens, one-click leads, company page clicks, date. The granularity of the data is (per memberId, advertiser Id). The datasets were removed shortly after data analysis to comply with data handling and retention standards. The data itself did not have any noise applied to it in its offline state for use in the data analysis. The structure of this dataset is showcased on LinkedIn's advertising platform, Campaign Manager. A representative day of data is illustrated in Table 2, aggregated by advertiser.

LinkedIn primarily uses this dataset to report campaign performance metrics via the Campaign Manager platform, and it has undergone years of scrutiny from engineers and users and describes the count of non-duplicate actions by the member. Campaign performance refers to a wide range of metrics for LinkedIn campaigns that measure ad performance. The dataset includes sixty-three numerical variables that describe user behavior towards advertisements, e.g.,

Table 2 Statistics of a single day for user engagements, aggregated by advertiser

Summary	Impressions	Clicks	Landing page clicks	Cost (USD)	Likes
Count	117K	117K	117K	117K	117K
Mean	6.4K	76	26	97	2
Std Dev	97K	5K	2.6K	1.8K	29
Max	21.7M	1.21M	872K	530K	7.6K

Summary	Comment likes	Comments	Shares	Follows	Video views	Company page clicks
Count	117K	117K	117K	117K	117K	117K
Mean	0.01	0.03	0.04	0.4	332	3
Std Dev	0.33	0.53	0.7	6.1	4.8K	69
Max	75	88	117	1.2K	809K	21.4K

impressions, clicks, and comments. It contains over a million member entries each day; over five-hundred thousand of those are unique. The study performs analysis on twelve months of data. In total, the study captures over 200-million-member advertisement interactions at the daily granularity.

From the data source, we select several literature-backed engagement proxies that record visits to advertiser company pages, interaction with advertisements, such as comments, likes and shares, and record advertiser cookies from landing page visits and product purchases. Table 3 provides an overview of the metrics used for the case study, including the considered timeframes and number of user behavior clusters.

6.2 Temporal granularity

Considering the timeframe, our study tracks longitudinal engagement from twelve-months of LinkedIn engagement data. Longitudinal engagement is the member's actions towards an advertiser over a given timespan. Because we aggregate data at the monthly granularity, longitudinal engagement requires further tracing of member engagement across the months as they change clusters/funnel stages. Therefore, the study also tracks the member-advertiser cluster and the summations of all engagements for each month. The collected data allow the analysis to study both the long-term behavior changes of members and the engagements between clusters. The input data for the subsequent K-Means clustering model consists of integers starting with zero. For example, ten impressions denote that the member viewed ten of an account's advertisements via LinkedIn advertising over one month. Zero implies that no engagements or impressions were recorded.

The data is constructed by randomly sampling from the active advertiser pool over the study window and then including members those advertisers targeted. All filtering comply with LinkedIn privacy guidelines. The sampling design mirrors how exposure occurs on the platform, frequently through advertiser defined targets. The sampling yields heterogeneous sets of (member, advertiser) pairs spanning multiple demographics, industry, verticals, and geographics that mitigates selection bias.

In addition, we consider data pre-processing steps to mitigate the impact of the metric's scaling on the resulting K-Means clustering. K-Means is well known to be susceptible to variables having different scales and tends to focus on variables with larger ranges of values. We considered both scaled and unscaled data for K-Means clustering and found that a scaled approach is preferrable as we can include more diverse metrics to measure different user behaviors. Specifically, in our case the metrics with the largest scale include impressions, comments, and shares, whereas metrics with smaller scales include clicks and landing page clicks. We considered two approaches: the first is to use the original data with different scales, and the second is to transform and normalize each metric to a unit scale centered around zero with a standard deviation of one. Given that impressions are often 100 times higher when compared to landing page clicks, the result of clustering without scaling are clusters based on a range of impressions with minimal or no consideration of other metrics. In contrast, scaling all metrics allows each metric to influence the marketing funnel model equally. Hence, we decided to scale all metrics for the clustering.

6.3 Cluster centers and marketing funnel stages

A critical aspect for modeling the market funnel via clustering is to decide how many distinct clusters, or funnel stages, we want to create. For K-Means, the parameter k refers to this number of clusters. While there are multiple methods to determine a good value for the parameter k, the optimal value balances its ability to fit the dataset with model simplicity. A good k value will produce a low model loss, which describes how well the clusters fit the data, while ensuring clusters are both dense and well-spaced between their centers (i.e., user behavior clusters are sufficiently distinct) (Delattre and Hansen 1980). If we choose a k-value that is too large, we get a lower model loss but a more complicated model that is difficult to understand due to distinguishing many different user behaviors. In our case, we investigated k-values between 3 and 30, and using the data-driven approach we determined that a k-value of 10 provides a good balance between the similarity of user behavior within a cluster (homogeneity), differences in behavior between clusters (heterogeneity), and complements silhouette consistency.

The K-Means model creates k cluster centers that describe member behaviors on LinkedIn. We describe each of the ten created clusters by the set of metrics that comprise the respective cluster center. Each cluster's primary metrics are displayed in Fig. 3c. Where clusters share primary metrics, the scale of the metrics distinguishes one cluster from

Table 3 Engagement metrics inputs for the machine learning models

Design parameter	Description
Selected metrics	On Platform: Impressions; Likes; Shares; Comments; Comment likes; Company page clicks; One-click lead form opens; One-click leads; Follows On Advertiser Site: Landing page clicks; Post-click conversions; Post-view conversions
Timeframe (window size)	Daily, Weekly, Monthly (selected for final model)
Number of funnel stages	10 (chosen to balance cluster count vs. model loss)

the other. The study works with these clusters to model the marketing funnel, where each cluster refers to a stage in the marketing funnel. Further, when a user's behavior classifies them into a new cluster, we consider that a stage transition in the marketing funnel and use it to track the movement of members throughout our model of the marketing funnel.

The number of clusters further simplifies when focusing on the top-three metrics per cluster. Some clusters share the same top-three metrics, which results in further coalescence of clusters and is a tradeoff to improve model interpretability. The study tracked multi-cluster funnel movements of length 2, 3, and 4 to demonstrate that users are making forward movements through the funnel, instead of randomly skipping between stages. Month-to-month χ^2 tests show that both stage distributions and member movements vary more than expected by chance, indicating that the data-derived funnel captures real, evolving behavior. The finding also agrees with existing data that members change their behavior in response to advertisements.

In Fig. 3c, we list the metrics that best characterize engagements for members in the corresponding cluster. While we chose k based on a balance of model loss and the complexity of the created marketing funnel, some clusters are similar in the engagements that characterize their cluster. For instance, we can see that Cluster 3 and Cluster 9 share the exact same cluster metrics but have different mean cluster centers. These clusters denote different (e.g., higher or lower) levels of the same engagements. Note that we show 10 clusters (0–9) but later graphs only show clusters 0–7. We denote cluster numbers by traffic volume, so the last two clusters had relatively less traffic and transitions and were therefore excluded from some of the predictive analyses. Fortunately, these metrics are captured in other clusters, e.g., Cluster 8 has the same metrics as Cluster 6 and Cluster 9 has the same metrics as Cluster 3.

Our cluster analysis produces a separation of user behavior among many engagements, and Fig. 3c displays the key metrics and the standard deviation of each cluster's key engagements. Each cluster consists of members demonstrating a significant increase in the key engagements. Each axis shows the key engagement's absolute standard deviation from the mean user engagement across all clusters. Clusters up and to the right represent engagements further from the norm, such as rarer engagements or a larger increase in the user's engagement. In contrast, clusters to the left and downward capture smaller nuances of behavior change or exhibit details about more common engagements. Even when sharing the same behaviors (e.g., Clusters 3 and 9), the amount of the key engagement differs substantially.

The K-Means model produces clusters with centers that denote disjoint regions of engagement. Each cluster center represents its combination of user engagements. Each center is also separated from other clusters in space. The clusters are sorted by their size, descending from the largest cluster (0) to the smallest cluster (9). It follows the four largest engagement clusters are zero through four. Cluster 0 denotes a lack of customer engagement while on the LinkedIn platform, such as when someone enters the site and leaves after checking notifications. Cluster 1 represents post likes, other engagements, and comments. Cluster 2 emphasizes post shares, Cluster 3 is conversions, and Cluster 4 has a larger number of impressions. The other clusters are smaller and therefore contribute less to the marketing funnel. The study's created clusters highlight engagement consisting of post likes, shares, impressions, and external website conversions.

Clusters might share the same engagement metrics at different levels. For example, the metric "other engagements" exists in Clusters 1 and 2. Other engagements describe other ways members can interact with advertisements. The metric plays a role in classifying the member into either cluster, but the metric is more or less important in each cluster. In turn, if metrics depend on themselves, i.e., a metric is more likely to occur if it occurred in the past, we expect some dependence between clusters. For example, a member who has commented on an advertiser's post is more likely to comment on future posts. The sharing of metrics between clusters might create a dependency between clusters, and we will further explore if these prior engagements can be used to predict cluster transitions in the next step.

The primary clusters in the marketing funnel do not focus on metrics like clicks, company page visits, or lead generation. A normal aspect of a customer's transition from awareness to knowledge gathering includes clicking on advertisements. Yet, our marketing funnel includes post interactions and the final website conversion. This may indicate limitations with our model and the complexity of metrics like landing page clicks. While Clusters 5 and 6 include landing page clicks and Cluster 8 includes company page visits, fewer customers get classified into these clusters. One explanation is that clicks play a lesser role in distinguishing between states of engagement for members. Still, metrics like post engagement and impressions better distinguish member states within the market funnel.

Finally, we considered if the derived clusters can be directly used as separate marketing funnel stages, or if we can combine some of the clusters into the same funnel stage due to the representation of qualitatively similar behavior. Here, we decided to aggregate clusters that are built on the same main engagement metric into the same funnel stage. Using this mapping of engagements to clusters (Fig. 3c), we also derive the simplified marketing funnel shown in Fig. 3b in Sect. 10.1.

6.4 Dataset granularity for clustering

We also explored the impact of modeling user behavior at the daily, weekly, and monthly time granularity. Each time granularity describes the window size on which we aggregate data. We aggregate data from the respective day in UTC. The daily granularity dataset contains aggregate member engagements for each day. The weekly granularity aggregates daily user engagements during a week, whereas the monthly granularity aggregates engagements over a time of thirty days. Each resulting row in our data describes the member's engagement with respect to each advertiser. For example, if a member views advertisements from seven advertisers, our dataset contains seven rows for this member: each row aggregating all the member's interaction with one of the seven advertisers. An example of the monthly data stored for a member is shown in Table 1 in the appendix.

When comparing the different time granularities, we need to account for their ability to capture behavior changes. On the one hand, smaller granularities are better able to capture minor changes in user behavior. For instance, daily granularities can capture the immediate effect of marketing on user behavior in the subsequent days. Such granularity suits the body of existing research, which is focused on user behaviors in the days following advertisements. In contrast, larger granularities are better at detecting longer-term shifts in user behavior and might be more suitable for a B2B marketing funnel, which is concerned with longer aspects of the user-advertiser relationship. Hence, there are tradeoffs to any granularity used by models of the marketing funnel.

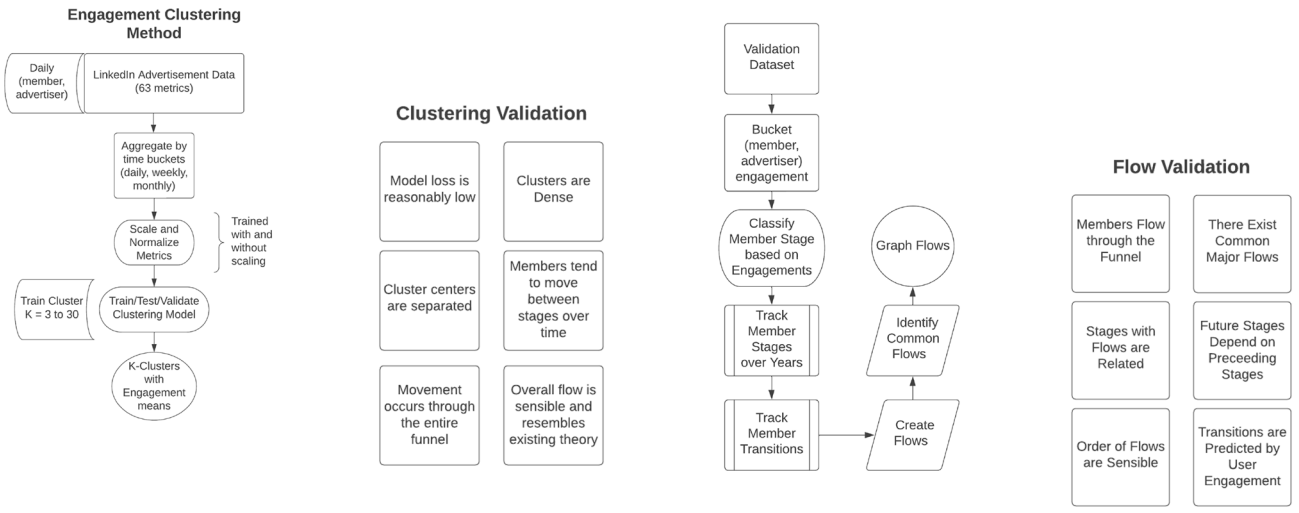
For our data, we identified that models trained on daily and weekly granularities struggle to produce stable cluster predictions. For example, datasets generated from daily and weekly granularities are sparse and tend to over-weight new engagements and result in less stable classifications. The marketing funnel generated from the sparse daily and weekly granularity dataset, hence, is less useful for capturing distinct and stable user behavior over time. Less stable predictions week over week make tracking user flows through the marketing funnel more difficult. In contrast, the monthly aggregated data capture larger, more distinct, and more stable changes in customer behavior toward an advertiser. Based on the monthly timeframe, we get several observations per member and twelve member classifications over the twelve-month timeframe of the study. We can then sketch these cluster classifications together to visualize their flow through the marketing funnel.

6.5 Model validation

We aggregate at advertiser level for stability, use well-known engagement metrics for interpretability, and verify (i) cluster compactness and separation, (ii) month-to-month mobility, and (iii) consistent explanatory power across three validation axes (see Fig. 2).

6.6 Appropriateness of the approach

To model the brand-member relationship, we aggregate data at the advertiser account level. This decision reflects the following methodological choice: rather than analyzing member interactions with isolated ad campaigns, we capture engagement with the brand as a whole. Since campaigns vary in content, timing, and creative design, aggregating at



(a) Process Flow for Engagement Clustering (b) Clustering Validation (c) Steps to Generate Stage Transitions (d) Flow Validation

Fig. 2 **a** Process flow for engagement clustering; **b** clustering validation; **c** steps to generate stage transitions; **d** flow validation

the brand level offers a more stable and comprehensive measure of longitudinal engagement. This enables us to model enduring member-brand relationships, which is essential for understanding behavior through the full marketing funnel.

The features used in our clustering and prediction models reflect key advertising engagement metrics (likes, shares, follows, comments, impressions, clicks) which are both behaviorally meaningful and industry-standard. These features not only enhance model interpretability but also align with how advertisers measure performance and ROI. By grounding the model in standard metrics, we ensure the resulting funnel structure is intuitive and directly actionable for practitioners. Additionally, this design allows our approach to be retroactively applied to historical campaign data to assess past performance and forecast future user actions.

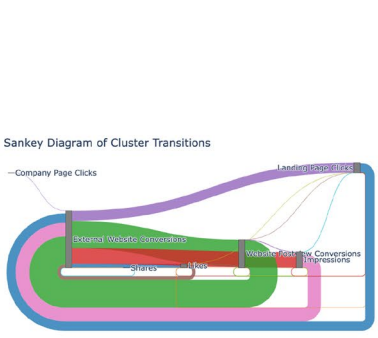
An unsupervised learning approach to uncover funnel stages directly from behavioral patterns is deployed. This avoids imposing a preconceived funnel structure and instead allows clusters to emerge from the data, offering a more authentic representation of user flow. By identifying patterns based on natural behavior rather than external definitions, we capture more realistic engagement dynamics. Figures 2c and d outline our procedure for modeling transitions and validating member movement, providing a reproducible pathway from raw data to interpretable funnel flows. To ensure methodological rigor, we validated our clustering model across three dimensions (Fig. 2b): model fit, cluster coherence, and explanatory power. For model fit, we optimized the K-Means clustering by selecting a *k*-value that minimized total loss while avoiding overfitting. We evaluated cluster coherence by confirming that cluster centers are both dense and distinct: clusters must meaningfully differentiate between user behaviors, enabling us to interpret each as a unique funnel stage. For example, members in two adjacent clusters may show similar types of engagement (e.g., likes and shares), but differ significantly in frequency or recency, justifying their separation.

A critical assumption of funnel models is that users should transition across stages over time. We tested this by analyzing monthly reassignments of members to clusters. Static classifications would suggest the model is not capturing meaningful behavioral change. Instead, we observe dynamic movement, with members flowing between stages month to month; evidence that the model captures genuine shifts in user behavior. These transitions were then analyzed to construct a simplified flow diagram (cf. Fig. 3b), validating that the emergent funnel structure matches expectations: initial exposure (impressions) leads to engagement (likes, shares), which in turn leads to deeper behaviors (comments, company page clicks, and conversions).

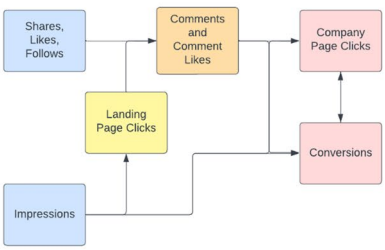
7 Flow validation

We generate common transitions by tracking member cluster classifications for each month. We generate monthly transitions by comparing the current cluster with the cluster membership in the subsequent month. A member who is classified into cluster 2 in June and then into cluster 3 in July will be marked as making a 2–3 cluster transition. Frequently, members undergo multiple months of being classified in the same cluster. The result of tracking all member transitions is that we can identify the most common transition for each cluster, which are noted as a common flow. Further, flows might also consist of multiple cluster transitions.

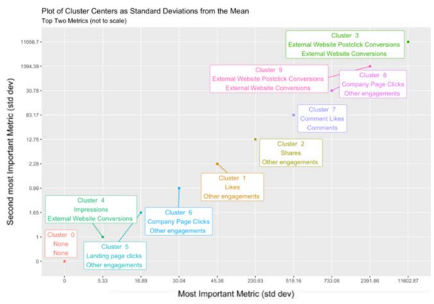
The study also tracks permutations of multi-cluster transitions. A cluster transition from 2–3 is a single transition, but many members make multiple cluster transitions over the duration of the study. For example, a member might proceed to transition from 2-3-5-1 over the course of a year. Our study tracks multi-cluster transitions to note common flows through the marketing funnel. Such flows represent common behavior shifts by members in the marketing funnel in response to advertisements.



(a) Cluster Transitions



(b) Simplified LinkedIn Marketing Funnel



(c) Metrics Characterizing Each Cluster Center

Fig. 3 **a** Cluster transitions. **b** Simplified linkedin marketing funnel. **c** Metrics characterizing each cluster center

8 Analyzing cluster transitions

We now focus on the analysis of the generated marketing funnel. We start with a descriptive understanding of the observed funnel stage transitions. The raw data contain stage membership for each member-advertiser combination per month. A cluster classification is the output from the K-Means model; an example is found in Table 2 in the appendix. A cluster transition describes changes in the member-advertiser's cluster from one month to the next. By recording cluster changes through transitions, our study is able to capture how member behavior is changing over time.

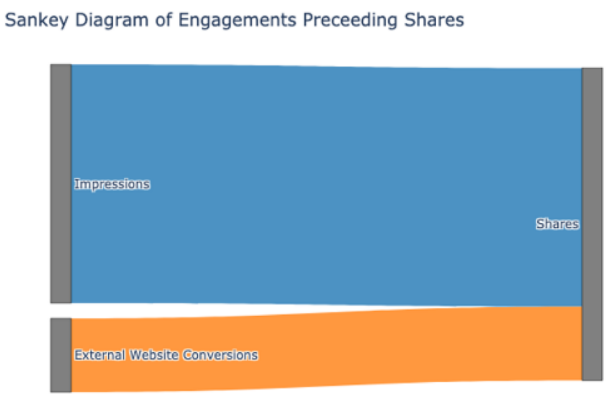
For each member, we record transitions through the funnel, i.e., from cluster x to cluster y, by calculating whether members at cluster x eventually reach cluster y. Recording multi-cluster movement allows the study to capture short and long-term transitions, simplifies data into a defined number of fields, and the compressed format is easier to process. Further, advertisers are ultimately interested in whether a member will eventually exhibit particular key behaviors, such as conversions. Therefore, our processing reflects desired information about eventual customer behavior in the marketing funnel. We capture both member cluster transitions overall, and whether members eventually visit particular clusters of interest.

Figure 3a provides a high-level depiction of important engagement changes as users move through the LinkedIn marketing funnel. We depict how users move between different funnel stages via Sankey diagrams, which display member transitions between clusters. Sankey diagrams allow for an intuitive representation of user flows between stages, and show greater volumes through thicker lines, i.e., thicker flows represent more frequent cluster transitions. Specifically, Fig. 3a depicts cluster transitions, where clusters (vertical grey lines) are represented via their most significant metric. The diagram shows a large volume of conversions, impressions, clicks, and company page clicks. Conversely, likes and shares occur to a lesser volume. However, instead of showing the cluster numbers from Fig. 3c, we choose the most significant metric in each cluster to make it easier to explain engagement changes.

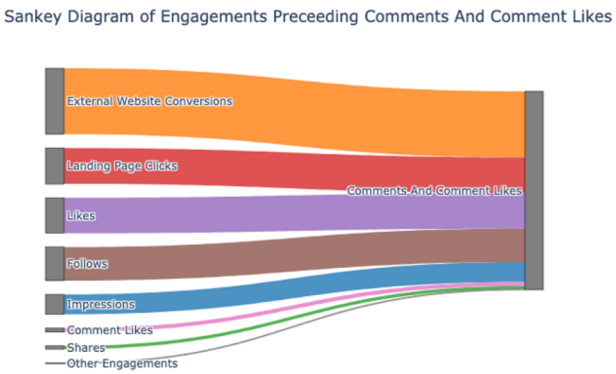
As mentioned before, in the Sankey diagrams we merge clusters with the same key engagement metric. For example, two separate clusters might both be related to liking posts, but one might have a higher mean engagement (higher number of posts). For the sake of simplicity, in the Sankey diagram we combine these clusters into a single place since our goal is to communicate the effect of that key engagement on other user behaviors.

9 Engagement influencer

The graphs in this section are derived from machine learning models which predict cluster transitions based on previous engagement. Specifically, we use a random forest model to predict the key engagement based on the engagements listed in Tables 2 and 3. The random forest model provides feature importance scores for its features, where a higher feature importance denotes a feature that is more important for predicting the key cluster transition. The below graphics display the feature importance for each model. For example, Fig. 4a displays the significant feature importance scores for predicting shares. In this case, we see that Impressions and External Website Conversions engagement types are significant for influencing future user behavior Shares. The engagements on the left side influence the engagement on the right. While the line thickness in the Sankey diagrams



(a) Impression and Conversions Are Predictive of Shares



(b) Multiple Prior Engagements are Associated With Subsequent Comments and Comment Likes

Fig. 4 Engagement influence examples: **a** impression and conversions generate shares; **b** many engagements affect comments and comment likes

before denotes increased visits, in the subsequent graphs in this section, line thickness represents feature importance from the random forest model. We now provide a short description of the key findings from these cluster transition prediction models.

First, we want to better understand which engagements are most important for predicting the transition to the cluster dominated by the Shares engagement type. Figure 4a shows that the most important features to predict Shares are Impressions and External Website Conversions. Users who see more impressions and convert are more likely to share advertiser content on social media. If users convert because they are happy with the product, they also tend to share it, which validates normative user behavior. Figure 4a provides confidence to advertisers that converting users will also share their content on LinkedIn.

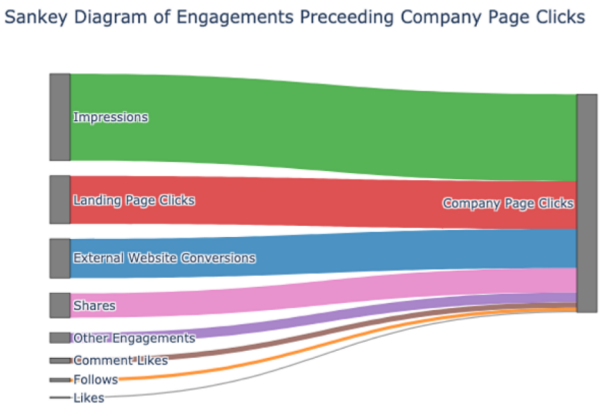
Figure 4b investigates users who comment or like advertisements. These users are already engaged with the brand. Following a brand has a more significant effect than impressions. Therefore, users already following the brand have an increased likelihood of commenting on their advertisements. It is worthwhile to note that users who comment on advertisements demonstrate high levels of engagement across many other engagement types such as landing page clicks, likes, and follows.

Figure 5a shows the importance of impressions in driving company page clicks for an advertiser. Otherwise, company page clicks are driven from high-cost engagements, such as conversions and landing page clicks. From the diagram, it seems that users who click to the company page exhibit late-stage engagements. In contrast, users who click to the company page are not highly engaging with the content from the company's advertisements, such as commenting on posts, sharing, or liking the content. Rather, advertisers drive less engaged customers to their company pages via impressions.

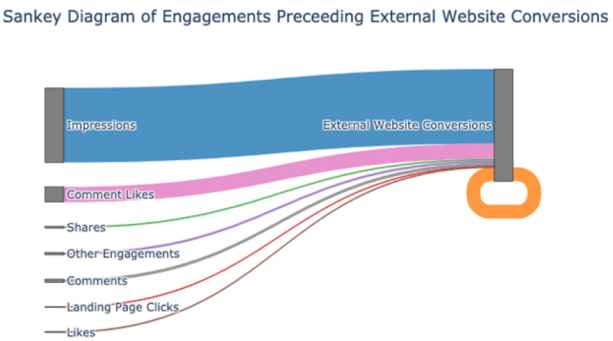
Many engagement factors play a role in external website conversions, as shown in Fig. 5b. Impressions play a large role in producing conversions. Surprisingly, comments and comment likes are also important factors that lead toward user conversion. Advertisers who are interested in producing conversions should focus on showing impressions and generating post likes and post comments. Further, customers who have previously converted for a company are more likely to convert again. We also see that many facets of user engagement with advertisement posts demonstrate small but positive effects on conversion.

Figure 6a depicts that users who have converted are more likely to see impressions on LinkedIn; consistent with re-targeting. Users who have converted are often re-targeted to further move along the funnel. Other engagements were not predictive of subsequent impressions in our models. Rather, the volume of impressions a user sees is a function of targeting on LinkedIn.

Members who click an advertiser's landing page often return and click again in later months, indicating recurring interest. Among engagement signals, impression volume shows the strongest association with subsequent landing-page clicks, while other on-platform engagements add comparatively modest predictive value. We also observe reciprocal associations: landing-page clicks are followed by later company-page clicks, comments, and comment likes.

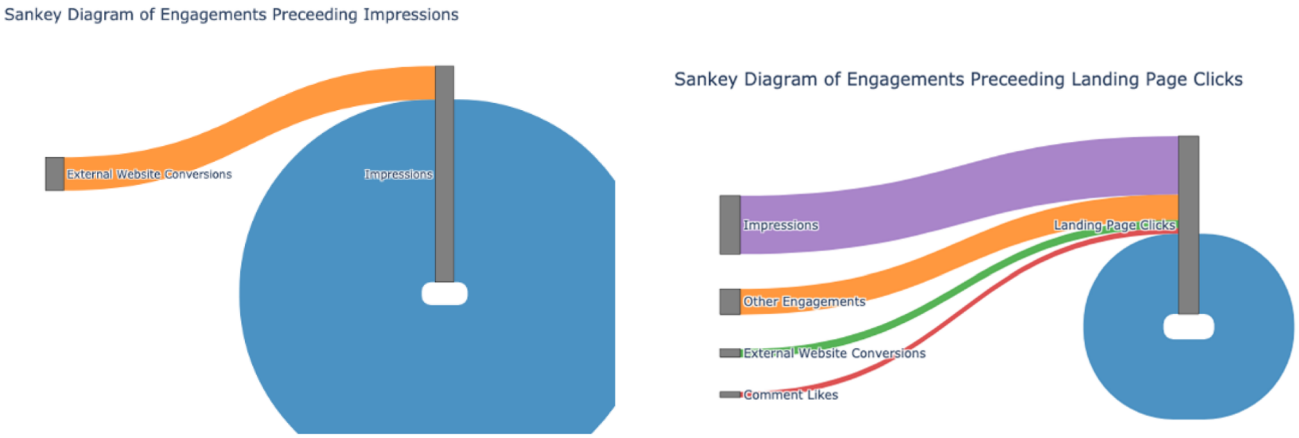


(a) Impressions, Clicks, Conversions, and Shares are Predictive of Company Page Clicks



(b) Impressions and Comment Likes are Strongly Associated with Conversions

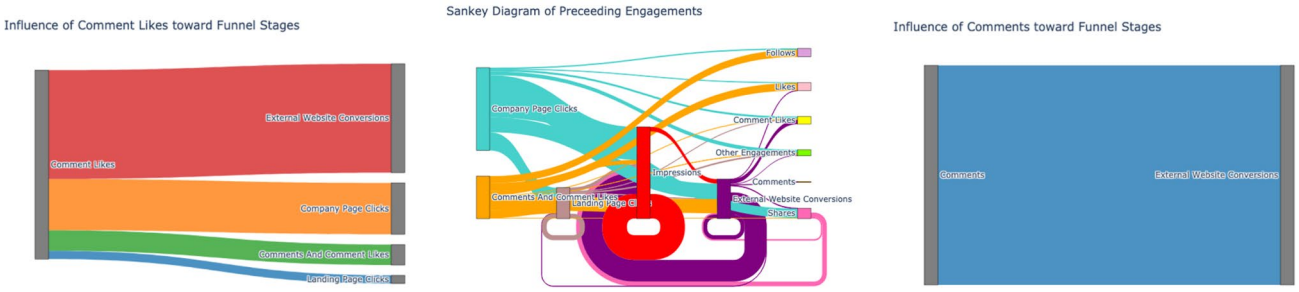
Fig. 5 Company page clicks and conversion influencers: **a** impressions, clicks, conversions, and shares affect company page clicks; **b** impressions and comment likes play a large role in conversions



(a) Prior Conversions are Associated with Higher Subsequent Impressions

(b) Prior Conversions are Associated with Higher Subsequent Impressions

Fig. 6 **a** Only previous conversions generate impressions; **b** Evidence of recurring interest: landing-page clicks over month



(a) Comment Likes are Predictive of Conversions and Company Page Clicks

(b) Engagement Influence Flows

(c) Comments are Predictive of Conversions

Fig. 7 **a** Comment likes influence conversions and company page clicks; **b** Overall engagement influence flows; **c** Comments as predictors of conversions

10 Predictor engagements

This section contrasts itself with the previous section; it visualizes the inverse data relationship, i.e., it considers the importance of single engagement types for generating other engagement types. These previous machine learning models predicted a key engagement based on the many engagements in Table 3 and the machine learning models provide a list of important features for each key engagement. This section flips that data and asks, across all the trained machine learning models, what was the role of engagement *x* across all key engagements? For this section, we compile the role of each engagement and predictive role, if any, on later engagement. Hence, this section shows a single engagement (on the left) and lists all the engagements it influences (on the right).

For example, in Fig. 7a we see in which predictive models the engagement type Comment Likes is an important predictor. Here, comment likes is significantly predictive of the random forest model predicting members transition to the external website conversion, company page click, comment and comment likes, and landing page clicks clusters. The size of the Sankey Diagram flows is the feature importance score from each separate random forest classifier. However, the feature importance scores are taken from multiple models. The line thickness describes the relative influence of the key engagement. Note that this section's line thickness is different than the Sankey Diagram in Fig. 3a, which denotes increased cluster visits. Researchers and practitioners can use the results of this section to understand how the engagement (left) influences future engagements in the marketing funnel (right).

Figure 7a tells the story about the powerful influence of comment likes on user behavior. Users who read

advertisement comments and like them are likely to convert, click to the company page, comment on the advertisement, and click to the landing page. Advertisers who generate interaction within the comments can expect to generate future engagement and even conversions from users.

Figure 7c shows how comments primarily influence future website conversions. Therefore, promoting engagement on advertisement posts via comments will influence future website conversions. We also see a synergy with the relationship from Fig. 5b where comments generate website conversions. Advertisers should take special interest in generating comment engagement for influencing future conversions.

Figure 8a depicts aspects of post-conversion brand loyalty and the likelihood of increased future engagement with posts. Note the inverse relationship where users who have converted are likely to interact with posts in the future, specifically via comments. We also see a strong relationship whereby users who convert tend to click to a company page; converted users later visiting company pages might depict loyalty to the brand.

Figure 8b shows how user engagement changes when they follow a brand. As expected, a user who follows a brand might click to the brand's company page in the future. More notable is that by following a brand there is an increased likelihood of commenting on brand posts, possibly because users are more likely to see posts from brands they have followed. The above figure shows a strong influence of follows on future comment likes. By encouraging follows brands can increase future post engagement which has future implications on user conversions.

Impressions are especially interesting since they are frequently driven by advertisers and targeting. In Fig. 8c we observe the positive effects of impressions on company page clicks and conversions. We also see a cyclical relationship in impressions where users are repeatedly shown impressions, likely because they are the target members for a campaign. Impression volume is less related to post engagement so advertisers should seek other means to drive increased post engagement. However, some small influence from impressions on post engagement does exist.

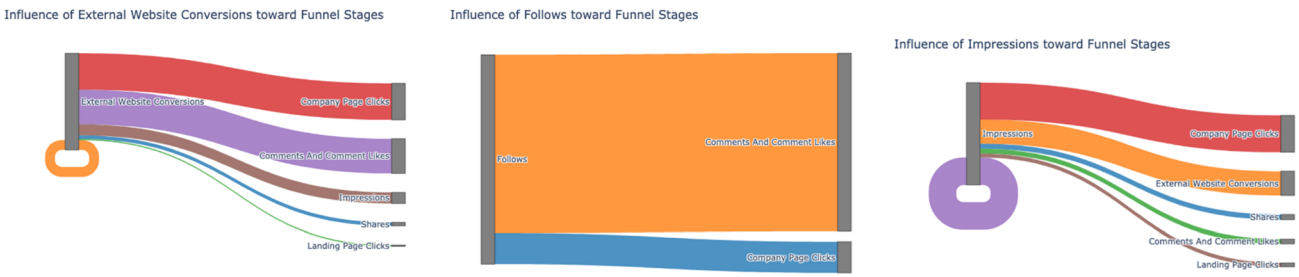
Figure 9a validates that users who tend to like are likely to make comments and like other comments. If advertisers can generate post likes, then they are likely to also generate comments and comment likes, each of which tends to produce conversions and company page clicks.

Figure 9b depicts the strong relationship between post sharing and company page clicks. Advertisers who wish to increase the volume of traffic toward their company page can do so via means whereby users share the advertisement.

The engagement type on the right is influenced by the engagement type on the left. For instance, the (far right) orange 'Comment and Comment Likes' can be interpreted as: 'Comment and Comment Likes' is influenced by Follows, Likes, External Website Conversions, and Landing Page Clicks. The right is the engagement being produced, and on the left are the engagements that occur in the preceding months.

10.1 Simplified funnel

To summarize the previous findings, Fig. 3b provides a simplified, high-level depiction of our use-case of the marketing funnel methodology for LinkedIn engagement data. The simplified marketing funnel describes the flow of members through the funnel and considers which engagements influence future behavior, and which behaviors proceed earlier engagements. Figure 3b describes which metrics are earlier marketing funnel metrics vs. later marketing funnel metrics. Due to its simplicity, it can easily be used to target specific behaviors and engagements.

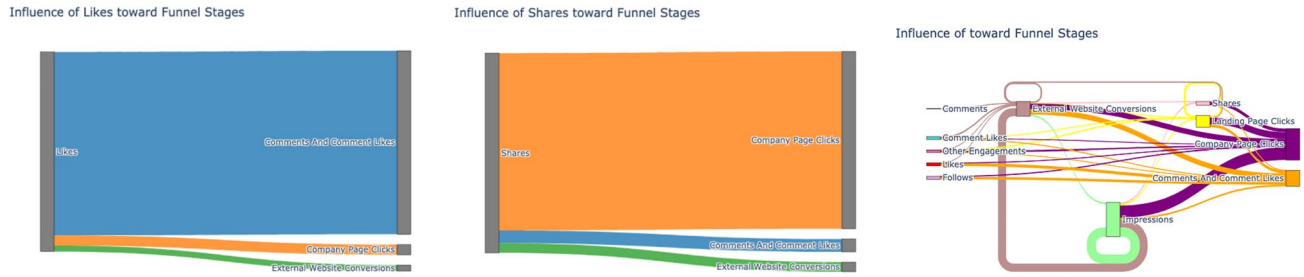


(a) Conversions are Closely Related to Company Page Clicks and Comments

(b) Company Follows are Associated with Higher Subsequent Post and Company Page Engagement

(c) Impressions are Related to Many Types of Engagements

Fig. 8 **a** Conversions vs. company page clicks and comments; **b** Impact of company follows on post/page engagement; **c** How impressions relate to various engagements



(a) Likes Influence Comment Likes

(b) Shares Influence Company Page Clicks

(c) Combined Influences from Right to Left

Fig. 9 **a** Likes influence comment likes; **b** Shares influence company page clicks; **c** Combined influences flow

11 Case study: predicting future funnel stage transitions

While the previous sections considered cluster transitions based on engagement types, we now focus on predicting cluster transitions based on user behavior. Here, we consider two different models:

- Transitions based on previous clusters: The first model predicts future cluster transitions as a function of historical clusters.
- Transitions based on previous engagements: The second model predicts cluster transition as a function of historical member engagements.

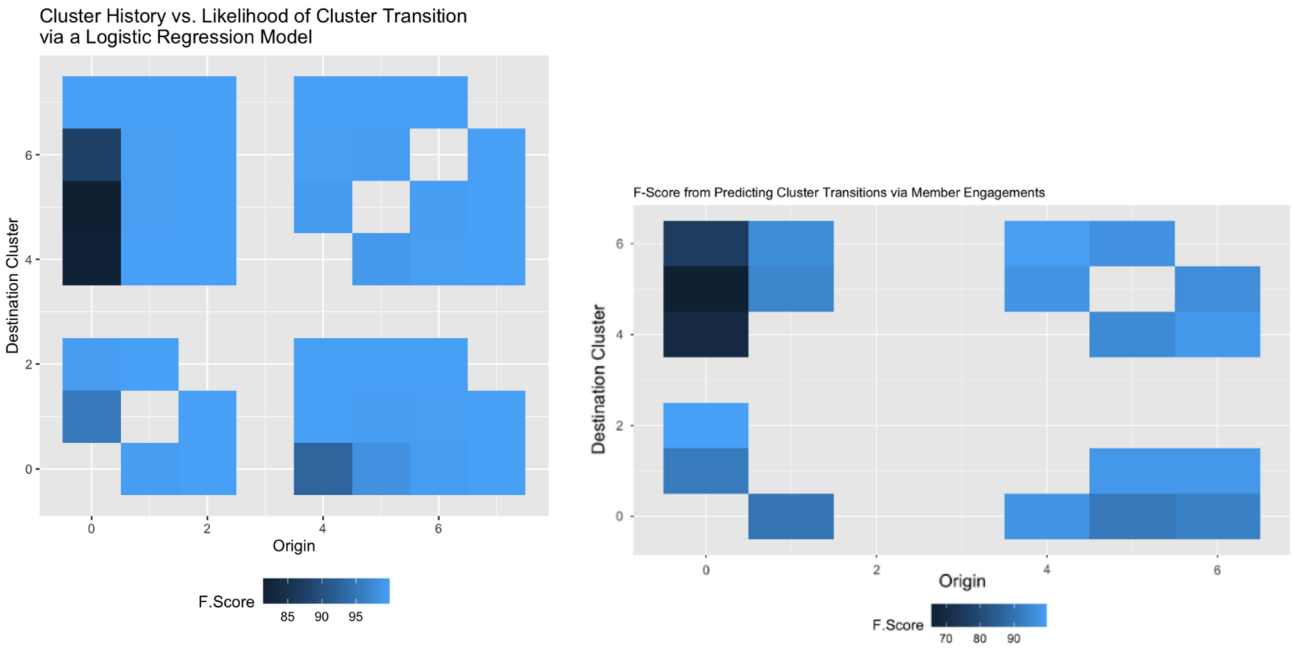
Each approach examines the relationship of a future cluster transition with a different variable: historical clusters vs. member engagements. The approaches capture different aspects of how cluster transitions interact with member behavior. The result of the approach outlines how cluster transitions are a function of both historical clusters and member engagements.

The first model's performance shows the predictive power that previous clusters have on a future cluster. A high-performance suggests that clusters are interrelated. A low performance might suggest cluster independence, or a lack of dependence on previous clusters, which might challenge traditional facets of marketing funnel theory. Therefore, we hope to validate existing marketing funnel theory by proving a strong predictive nature of past and future clusters.

The second model's performance is indicative of how much previous engagements can predict future cluster stage transitions. Integrating engagements can be challenging since most members undergo months of reduced advertiser engagement. One of our primary clusters represents baseline member engagement (therefore, cluster 0 has no 'key' engagements). On the one hand, it is common for members to exhibit periods of reduced engagement. Such periods might be due to platform frequency caps that reduce the number of impressions members see each month. Members might also not use LinkedIn on a regular basis. Models of the market funnel should handle situations where member engagement temporarily decreases with an advertiser.

Our study shows that LinkedIn effectively reaches members across a wide range of brands, even if not every impression leads to an immediate interaction. Many members consistently see ads, creating ongoing brand awareness and the opportunity for future engagement. For example, a member who sees one ad from each of twelve brands every month for a year contributes to valuable brand exposure even if they only interact with select subset of advertisers. To illustrate further: if this member instead sees twelve ads per month, interacts with one, and repeats this pattern over four months, our dataset would record four active classifications, which amounts to about 3% of total (member, advertiser) combinations while still showing regular ad exposure. While this leads to sparsity in interaction data, the size and richness of our dataset allow us to capture a broad range of member-advertiser transitions. We see a wide variety of these transitions, giving us confidence that the model reflects diverse member behaviors and meaningful shifts in engagement patterns over time.

In the next section we train two classification models. In the first model, we explore the ability of a logistic regression model to predict the future cluster based on historical clusters. In the second model, we train a random forest classifier to predict cluster transitions based on member engagement. The logistic regression provides log odds for the importance of cluster history, which gives us statistical significance. The random forest classifier provides feature importance scores for each engagement, which describe how important each engagement is for predicting each permutation of cluster transition. Each of the models provides us feedback about how the marketing funnel can inform us about future user behavior.



(a) F1 Scores for Predicting Cluster Transitions based on Previous Cluster History

(b) Predicting Cluster Transitions with Member Engagements

Fig. 10 **a** F1 scores for transitions based on previous cluster history; **b** Predictive performance of member-engagement-based models for cluster transitions

11.1 First model: predicting cluster transitions from historical cluster membership

The first predictive model examines the effect that historical clusters have on cluster transitions using a logistic regression model. Historical clusters are those previously visited by a member. The target classification is a Boolean variable that indicates whether a member eventually transitions to the target cluster. We create a model for each target cluster. The model takes historical cluster classifications as input and predicts the likelihood of a member visiting the target cluster. A performant model indicates a relationship between past and future clusters.

The model is based on the theory that members tend to move from more shallow initial stages within a marketing funnel toward later stages. If members tend to move from shallow to deeper stages, then a strong predictive relationship exists between stages. By creating a model for each target stage, we evaluate its connection with other stages. By using logistic regression, we can provide simple coefficients that capture the strength of relationships between historical stages and the target stage. While we could improve performance through more advanced models, our goal was to validate the strength of stage interdependence.

For the first model, the input is the information about advertiser-member's historical clusters. For example, in a model with ten clusters, the model takes the other nine cluster columns as input. To prepare the further analysis of user/member transition between funnel stages, we also add binary columns to each member/advertiser record to specify whether the member has previously visited the cluster. The binary features are input into the subsequent predictive analytics models which predict if the member will eventually visit the target cluster. The model predicts the probability of the cluster transition based on the historical clusters. We show the performance of each model in Fig. 10a, which captures the correlation coefficient of each cluster on predicting the target cluster. Blue spots denote performant models with high F1-Scores (indicating high values for both Precision and Recall), and gray spots depict poor and non-significant model performance. Figure 10a shows that we can predict most user cluster transitions based on historical clusters.

The logistic regression models predict whether the member will ever move into the target cluster and correctly predict this movement to the future cluster at least eighty percent of the time. Each logistic regression model shown in blue has an F1-score over 94% and a high true positive rate despite many negatives in the sample. The study establishes a relationship between clusters where knowledge about other cluster movements can predict whether the member will ever move into other clusters.

The success at predicting future cluster performance is somewhat surprising since there is no correlation in cluster occurrence. Correlation coefficients are near zero between

all clusters. However, the model has great performance in predicting if a member will visit the cluster in the future. Hence, while no direct correlation exists between clusters, the findings indicate that dependencies exist among clusters.

11.2 Second model: predicting cluster transitions from user engagements

The second approach examines the effect of user engagements on cluster transitions. User engagements are numerical metrics, such as impressions and landing page clicks, which describe the interaction between a member and a brand's advertisements. If the member had no engagement these metrics are zero and might be as high as fifteen per campaign per month. We expect members who engage with ads to move deeper into the marketing funnel. Therefore, we expect a model based on member engagements to be able to predict marketing funnel transitions.

The study seeks to understand which metrics are predictive of cluster transitions. Predictive metrics are those that indicate an increased likelihood of cluster transition. If advertisers use these metrics, they can better identify members ready to exhibit the cluster's key engagements. In this second predictive analysis, the models use a random forest classifier. Similar to before, we provide F1-Scores of these classifiers for the different models that predict the target cluster based on historical user engagements. Model performance varied widely for predicting cluster changes. Figure 10b shows model performances and Table 3 (in the appendix) lists important model features. Note that Table 3 only lists the selected important metrics, while the long list of tested metrics is shown for the LinkedIn AdAnalyticsV2 statistics API.

Figure 10b visualizes the success of predicting cluster transitions based on historical user engagements. Blank table entries denote either non-existent transitions, such as 0-0, 1-1, 2-2, 3-3, etc., or that models produced an F-score below 65% when predicting cluster transition. We can see that many cluster transitions are difficult to predict. However, the models with high F1-Scores demonstrate that some cluster transitions are relatively accurate to predict using previous member engagements. The random forest classifier model can predict member transitions to the non-engagement cluster with an F1-score over 90%, which is sensible since many members eventually have months of minimal activity. The highest performing models achieve a 50–70% specificity for predicting cluster for conversions and 10–15% for cluster corresponding to landing page clicks. The engagement trained model can predict clicks and product conversions but struggle to predict other cluster transitions.

We exclude the metrics that best describe the target cluster from the model since these metrics have a strong relationship to the target cluster. Instead, our goal is to capture the ability of other metrics to predict the target cluster. Our exclusion of metrics highly correlated with the target cluster reduces our model performance to a degree but provides better nuance and increased insight in how other engagements can predict the desired critical member behavior and its corresponding cluster.

As a robustness check, we also trained logistic regression models for the different cluster transitions. However, since the random forest classifier outperforms the logistic regression models, we do not show the logistic regression model separately. When comparing the logistic regression and random forest models, the random forest classifier performs better on a larger number of metrics, precisely when predicting transitions to stages that correspond to conversions and landing page clicks. This indicates that conversions and landing page clicks might be more difficult to predict, or that including interactions of engagements, which the random forest classifier can learn, are helpful for the prediction of future cluster transitions.

The paper provides both logarithmic and random forest classifier models for each stage transition. As a result, there are many models. For the sake of space, it is useful to provide heat map graphic with the F1 score than individual ROC graphs within the paper. The F1 score is at the highest balance of precision and recall for each model. The research inspected ROC curves for each model as a sanity check and observed no unexpected behavior. This study emphasized whether models can predict future user behavior and does not delve deeply into the nuances of those models. The study kept diagnostics concise and focused on F1, noting that stronger learners (e.g., gradient boosting or DNNs) could improve accuracy but are outside scope.

11.3 Linking empirical results to research gaps

This study seeks to address a specific gap for positioning the marketing funnel's structure and flow: most frameworks rely on predefined, theoretical or business-specific, coarse stage and flow definitions that offer limited guidance on how behavior-derived stages relate to prospective movement through the funnel. The empirical results from this study link back to this gap in three ways. 1. Behavior-derived stage structure, 2. Prospective transition prediction from stage history, and 3. Antecedent engagement as signals of movement.

Behavior-derived stage structure provides clustering results that yield interpretable engagement-based stages rather than imposed labels. Flow analyses show ordered movement between these stages rather than random

hopping, indicating that learned states capture stable behavior regimes. This addresses the need for stage definitions grounded in observed patterns rather than assumptions. Prospective transition predictions from stage history achieve strong sample performance and demonstrate that movement between the behavior-derived stages is prospectively predictable. This connects learned state structures to forward-looking dynamics and close the gap between descriptive stage maps and actionable transition forecasts. Antecedent engagements as signals of movement identify which interactions act as leading indicators of deeper behavior (e.g., the role of comments and comment likes before conversion, follows before downstream post activity, impressions before company-page visits). This links micro-events to upcoming transitions and clarifies when and how specific engagements signal imminent movement.

The results generate stable monthly classification while still capturing monthly movements. This clarifies where the approach is the most informative over the longer horizon rather than at a high-frequency or single-day effect and provide practice framing use. Taken together, the results provide a reusable method to derive funnel stages directly from behavior, show that transitions are prospectively predictable, and map the engagements to subsequent movements. The discussion connects empirical findings to the stated gap and delineate the paper's contribution as behavior-grounded.

12 Discussion and conclusions

This paper presents a novel, data-driven methodology for modeling the marketing funnel using proprietary behavioral data from LinkedIn. Unlike traditional funnel models based on limited or aggregated marketing data, our approach leverages detailed user engagement metrics to infer customer stages and transitions. This work contributes to the growing field of computational social science by combining social network analysis, behavioral modeling, and predictive analytics in a real-world digital platform context.

Specifically, our analysis reveals nuanced behavioral insights that challenge some assumptions in marketing analytics:

- **Late-Stage Indicators:** We identify comments and comment likes as key latent indicators of imminent conversion. These interactions-overlooked in many models-are among the most predictive behaviors prior to conversion, showing that deeper social interactions carry greater intent signals.
- **Early-Stage Engagements:** Likes, shares, and follows are foundational behaviors. They initiate user-brand interaction and serve as early predictors of movement into deeper funnel stages. These behaviors also play a role in social contagion and visibility within the network.
- **Landing Page Clicks as Mid-Funnel:** Often treated as a key conversion metric, we find landing page clicks to be transitional. They do not independently predict conversions but are intermediaries between awareness and deeper interactions.
- **Company Page Clicks and Impressions:** Surprisingly, company page clicks emerge as a late-funnel behavior with strong ties to conversion. Meanwhile, impressions-while not directly influential-facilitate transitions to mid- and late-stage behaviors, highlighting their indirect yet essential role in user journey progression.

These findings demonstrate that different engagement types function differently depending on their position in the behavioral sequence, and that a nuanced understanding of such sequences can guide strategic marketing interventions.

12.1 Modeling customer behavior within a social network

Conventional models treat the marketing funnel as a static or opaque construct. In contrast, we empirically derive its structure using clustering techniques applied to fine-grained LinkedIn user engagement data, capturing distinct stages and the flows between them. Our findings show that user behavior changes over time are not merely stochastic fluctuations, but meaningful transitions that reflect deeper engagement and movement through the funnel. The high F1-scores of our predictive models confirm that the clustering represents more than random behavioral patterns-it captures evolving user states in a socially networked context. Moreover, we provide evidence that member engagement is path-dependent: past behaviors, such as liking or sharing, influence the probability of deeper interactions such as commenting or clicking on company pages. This interdependence between engagement types reflects latent behavioral structures within the user network and supports the idea that social ties and platform interactions can be mined to understand conversion likelihood.

12.2 Temporal granularity and predictive dynamics

We find that temporal aggregation significantly affects the interpretation and predictive power of behavioral data. Daily aggregation improves sensitivity to short-term behavioral shifts, useful for predicting immediate transitions (e.g., sudden interest in a product), while monthly aggregation better captures long-term trends (e.g., gradual brand affinity). By demonstrating that different time granularities expose distinct facets of customer behavior, we propose that

researchers and practitioners should construct multi-temporal models for more comprehensive behavioral analysis, an important insight for temporal network mining and predictive social modeling.

12.3 Methodological contributions

This study offers several methodological innovations that advance the understanding of user behavior within networked systems. First, we operationalize the marketing funnel as a dynamic, cluster-based behavioral model. By applying unsupervised learning (K-Means clustering) to engagement data, we define funnel stages empirically, enabling a data-driven view of user progression through behavioral states rather than relying on pre-defined or assumed structures. This approach provides a flexible framework that can adapt to different contexts and engagement types.

Second, we apply supervised machine learning models to predict transitions between behavioral clusters. This predictive layer allows us to identify which users are likely to progress toward deeper engagement, based on their historical behaviors. The ability to anticipate such transitions at scale represents a powerful tool for both researchers and practitioners seeking to understand the evolution of user behavior in social networks.

Third, our analysis uncovers latent structures in user engagement types, identifying which behaviors act as precursors to key funnel transitions. By examining feature importance and model performance across clusters, we reveal how different engagement metrics contribute differently depending on the user's current stage. These insights not only enhance the interpretability of the funnel model but also provide a framework for evaluating engagement strategies based on behavioral impact.

Collectively, these methodological contributions enable a scalable, customizable, and empirically grounded approach to modeling and analyzing marketing funnels in digital and social platforms. The approach is extensible across platforms with similar engagement data, offering value to both academic researchers in social network analysis and practitioners working in data-driven marketing and user experience optimization.

12.4 Limitations, platform specificity, and generalizability

We acknowledge that the study's platform-specific nature. Drawing on LinkedIn's unique engagement types such as company follows and job-related content limits immediate generalizability. Engagement behavior may differ across platforms with different social norms and user goals (e.g., Facebook or TikTok), which might produce different funnel structures. In addition, the use of K-Means clustering and fixed hyperparameters introduces some sensitivity to model design decisions. However, this design flexibility is a strength, as it allows practitioners to adapt the approach to specific goals and datasets. Further, our study models a single-user-to-brand relationship. In reality, users interact with multiple brands, and cross-brand influence could impact funnel progression. Future work should explore such networked, multi-brand interactions and extend the model to capture broader ecosystem dynamics.

The analysis relies on proprietary data that cannot be publicly released and no longer exists due to data handling compliance. While variables and procedures are described in detail, full replication is constrained by access restrictions; reproducibility is best interpreted as procedural rather than data replication.

There is potential survivorship selection bias in advertiser participation. Advertisers that run campaigns on LinkedIn might systematically differ systematically from those that do not. Further, member targeting creates non-random exposure across members. These factors can shape the composition and movement within the derived stages. The results should be interpreted as observational and predictive rather than causal.

13 Conclusion

By linking behavioral data to measurable stages of customer interaction, our study builds a bridge between theoretical marketing constructs and empirical network-based analysis. The methodology and findings offer actionable insights for researchers and practitioners alike, providing a roadmap for mining social platforms to understand and influence customer journeys. Our work illustrates how social network platforms like LinkedIn can serve as powerful laboratories for behavioral modeling, enabling deeper exploration of the structures and dynamics underlying digital customer engagement.

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Author contributions Conceptualization: All (Defined the research goals, formulated the problem, and designed the overall methodology.) Methodology: CC, CH (Developed the clustering and predictive modeling framework, including data aggregation and validation strategy.) Data Curation: CC (Processed and aggregated LinkedIn engagement data; managed data quality and platform-specific preprocessing.) Formal Analysis: CC (Conducted machine learning analysis, cluster validation, and interpretation of funnel transitions.) Software: CC (Implemented the modeling pipeline and analytics scripts used in the study.) Visualization: CC (Created figures for clustering validation, user flows, and funnel transitions.) Investigation: All (Synthesized practical

implications from data findings; analyzed marketing funnel stages and strategic insights.) Writing – Original Draft: CC, MH (Led the drafting of the manuscript, including framing of contributions and discussion.) Writing – Review & Editing: MH, CC (Provided critical feedback and revisions to improve clarity, argumentation, and technical accuracy.) Supervision: MH, CH (Oversaw the research process, provided strategic guidance, and ensured alignment with research objectives.) Project Administration: MH, CC (Coordinated tasks, timelines, and communication across the research team.)

Data availability The data that support the findings of this study are proprietary to LinkedIn, which limits the availability of these data, and so are not publicly available.

Declarations

Conflict of interest The authors declare no Conflict of interest.

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